
Real vs AI-Generated Face Detection and Interpretability

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Abstract

In this project, we study the problem of detecting AI-generated faces using a CNN (ResNet18) and a Vision Transformer (ViT). While both models achieve strong classification performance, they rely on different features: CNNs focus on local texture patterns, whereas transformers capture more global structure. Using Grad-CAM, we analyze how each model makes decisions and show that they exhibit distinct behaviors. These results highlight the importance of interpretability and demonstrate how model architecture influences deepfake detection.

1 Introduction

The rapid advancement of generative models such as Generative Adversarial Networks (GANs), diffusion models, and face manipulation techniques has made it increasingly difficult to distinguish between real and synthetic images. AI-generated faces are now highly realistic, raising serious concerns regarding misinformation, identity fraud, and digital trust.

Recent work has shown that deep learning models, particularly convolutional neural networks (CNNs), can effectively detect synthetic images by learning subtle visual artifacts. However, most existing approaches focus primarily on improving classification accuracy, with limited emphasis on understanding how these models make decisions. This lack of interpretability makes it difficult to assess whether models rely on meaningful visual cues or spurious correlations.

In this work, we address both detection and interpretability. We train a ResNet-based classifier as well as a Vision Transformer (ViT) to distinguish between real and AI-generated faces and further analyze their decision-making process using Grad-CAM. Unlike prior work, we investigate how the models' attention varies across different types of fake images, specifically GAN-generated, diffusion-generated, and FaceSwap images. Our goal is to provide insights into whether different generation methods lead to distinct attention patterns.

We show that while the models achieve strong classification performance, they also exhibit different attention behaviors depending on the source of the image. This highlights that deep learning models adapt their feature extraction strategies based on the characteristics of the generation process.

2 Related Work

Detecting manipulated and AI-generated faces has been widely studied in recent years. Rössler et al. [4] introduced the FaceForensics++ dataset, which provided a large benchmark for evaluating deepfake detection methods under different levels of compression and manipulation techniques. Their work showed that while deep learning models can perform well on clean data, they often struggle when image quality is degraded, suggesting that models may rely on superficial visual artifacts. This motivates our focus on understanding what visual cues detection models actually use.

On the modeling side, Dosovitskiy et al. [3] proposed the Vision Transformer (ViT), which applies self-attention mechanisms to image patches and captures global relationships across the image. In contrast, convolutional neural networks process images through local filters and hierarchical feature extraction. Geirhos et al. [2] demonstrated that CNNs are biased toward texture rather than overall shape, meaning they may prioritize fine-grained surface patterns over global structure. This distinction is particularly relevant for AI-generated face detection, where synthetic images may contain subtle texture inconsistencies but realistic global geometry.

To better understand model decisions, interpretability methods have become increasingly important. Grad-CAM, introduced by Selvaraju et al. [5], provides a way to visualize which regions of an image contribute most to a model’s prediction by leveraging gradient information. While originally designed for convolutional networks, similar ideas have been adapted to transformer-based architectures, enabling comparison of attention patterns across different model types.

In addition to architecture differences, recent studies such as Wang et al. [6] have shown that convolutional networks can successfully detect GAN-generated images by identifying statistical artifacts, although these artifacts may diminish as generative models improve. This highlights the importance of understanding not only whether a model works, but how it works and what features it relies on.

As for vision transformers, prior work has shown that they differ fundamentally from CNNs in how they aggregate spatial information. Raghu et al. [9] analyzed self-attention distances and found that even in the earliest ViT layers, self-attention heads consist of a mixture of both local heads and global heads, enabling the model to capture both nearby and long-range relationships from the beginning. In contrast, CNNs are structurally constrained by fixed convolutional kernels and therefore attend primarily to local features in lower layers, gradually building global representations only in deeper layers. Furthermore, at higher ViT layers, nearly all self-attention heads become global, reinforcing the model’s ability to capture long-range dependencies. This behavior is also consistent with the original Vision Transformer architecture proposed by Dosovitskiy et al. [10].

In this project, we build on these works by directly comparing a CNN-based model and a Vision Transformer on the same task of AI-generated face detection. Unlike prior work that focuses primarily on classification performance, we emphasize interpretability by applying Grad-CAM to both architectures and analyzing whether they rely on different types of visual evidence.

3 Methods

3.1 Overview

In this project, we aim to distinguish between real and AI-generated faces using deep learning models. We implement and compare two architectures: a convolutional neural network (ResNet18) and a Vision Transformer (ViT). In addition to classification performance, we analyze model behavior using Grad-CAM visualizations to understand where each model focuses when making predictions. Not only do we compare Grad-CAM results on real vs fake face images, but we also take it one step further in analyzing how these results compare across different types of fake images: GAN, Diffusion, and Face Swaps.

Our approach consists of three main components:

- Training a CNN-based classifier (ResNet18)
- Training a transformer-based classifier (ViT)
- Interpreting both models using attention and activation visualization

3.2 Intuition and Motivation

The baseline for this project is training a convolutional neural network and a vision transformer classifier on real vs AI-generated faces, achieving high accuracies. However, we are also interested in understanding why each model makes these decisions. This motivates the use of Grad-CAM to analyse the regions of the image that help these models make their predictions.

The key idea is that:

- ResNet focuses on local spatial patterns and hierarchical features

- ViT captures long-range dependencies and global structure

CNNs process images using local receptive fields and hierarchical feature extraction, and are effective at capturing spatial patterns such as edges, textures, and shapes. Which is why we hypothesize that the CNN may focus more on localized artifacts, whereas the Vision Transformers may better detect global inconsistencies, by modeling global relationships between image patches using self-attention mechanisms.

In addition to understanding how each of these models make their decisions in classifying real vs fake face images, we also take advantage of the structure of our dataset which gives us categorized fake images into: GAN, Diffusion, and FaceSwaps. This allows us to analyze interpretability across different kinds of AI-generated face images. We hypothesize that GAN and FaceSwap images will produce stronger localized responses, as they contain local artifacts that align with CNN behavior. Diffusion images on the other hand will produce weaker, more distributed responses, as they are more globally consistent, aligning better with ViT behavior.

3.3 Dataset Structure and Category-Based Analysis

The dataset used in this project from Kaggle [7] contains two complementary directory structures that enable both training and detailed analysis. It has a folder dedicated only for real and fake images, and another one that extends the fake images into subsections: GAN, Diffusion, and FaceSwap. Since the naming and overall structure of the folder from Kaggle is confusing and hard to navigate, We have rearranged it in the following way:

The models are trained only on the binary labels from the dataset/ folder. The source/ folder (which contains the categorized fake images) is used exclusively for Grad-CAM analysis. This allows us to evaluate how the models behave across different types of synthetic images without explicitly training on these subcategories.

3.4 Model Architectures and Training Setup

We compare two architectures for deepfake detection: a convolutional neural network (ResNet18) and a transformer-based model (ViT-B/16).

ResNet18 is initialized with ImageNet pretrained weights and adapted for binary classification by replacing the final fully connected layer. It primarily captures local spatial features through convolutional layers.

ViT-B/16 splits each image into 16×16 patches and processes them using transformer encoders. The classification head is replaced with a custom structure consisting of a linear layer, GELU activation, dropout, and a final linear layer for binary output. Unlike CNNs, ViT captures global relationships between image regions.

Data augmentation (horizontal flip, rotation, color jitter) is applied during training for both models. The best models are selected based on validation performance.

Parameter	ResNet18 (CNN)	ViT-B/16
Pretraining	ImageNet	ImageNet
Loss Function	CrossEntropyLoss	CrossEntropyLoss
Optimizer	AdamW	AdamW
Learning Rate	5×10^{-5}	1×10^{-4}
Weight Decay	1×10^{-4}	1×10^{-5}
Scheduler	Cosine Annealing	Cosine Annealing
Epochs	20	20 (Early Stopping)
Patch Size	N/A	16×16
Architecture Type	CNN (local features)	Transformer (global features)

Table 1: Comparison of training configurations for ResNet and ViT models

3.5 Model Interpretation with Grad-CAM and Attention Rollout

To interpret model predictions, we analyze both the CNN and Vision Transformer (ViT) using visualization methods that highlight which image regions contribute most to classification decisions. Our analysis compares real images and three types of synthetic images: GAN-generated, diffusion-generated, and face-swapped images. For both architectures, we examine individual predictions as well as category-level patterns. In addition to sample-level visualizations, we compute average activation maps across multiple images within each category to obtain more stable and representative patterns. We also generate difference maps (fake minus real), where positive values indicate stronger activation for fake images and negative values indicate stronger activation for real images. These maps help identify regions where the models respond differently depending on image type.

3.5.1 ResNet Grad-CAM

For the CNN model, Grad-CAM is applied to multiple convolutional layers (layer1 through layer4) to examine how model attention evolves across different levels of feature abstraction. Earlier layers capture low-level visual patterns, while deeper layers focus on higher-level semantic features. By comparing activation maps across layers, we analyze how the spatial distribution of important regions changes throughout the network.

3.5.2 Vision Transformer Grad-CAM

For the Vision Transformer, Grad-CAM is adapted to operate on patch tokens rather than convolutional feature maps. Gradients are computed with respect to the patch embeddings from the final transformer encoder block. The class token is removed, and the remaining patch tokens are reshaped into a spatial grid and upsampled to match the input image resolution. This produces class-specific activation maps that allow direct comparison with CNN-based Grad-CAM results while reflecting the token-based representation used by transformer architectures.

3.6 Vision Transformer Attention Rollout

To further interpret the ViT, we use Attention Rollout to visualize how information flows through its multi-head self-attention layers. Unlike Grad-CAM, which is gradient-based and class-specific, Attention Rollout relies only on the model’s learned attention weights and is therefore class-agnostic. Following the method introduced by Abnar and Zuidema [9], attention matrices are recursively multiplied across layers to estimate how the CLS token aggregates information from all image patches. This provides a structurally faithful view of how the final representation is formed and helps identify which regions contribute most strongly to classification decisions.

4 Experiments

4.1 Experimental Setup

Our experiments were designed to answer the following questions:

- How well does each model perform on the binary real-vs-fake classification task?
- How stable is the training process for each model?
- What visual cues does each model rely on when making predictions?
- Do these visual cues differ across fake image categories such as GAN, diffusion, and face swap?

To answer these questions, we evaluate each model using standard classification metrics and visual analysis. We then compare both models in terms of performance and interpretability.

4.2 Models Performances

The following table shows the performance metrics and results we got for the ResNet8 and the ViT models.

Metric	ResNet18 (CNN)	ViT-B/16
Training Accuracy	99.85%	97.97%
Testing Accuracy	96.84%	97.97%
Test Precision	99.15%	98.00%
Test Recall	97.61%	97.97%
Test F1-Score	98.37%	97.96%

Table 2: Comparison of training configurations for ResNet and ViT models

4.2.1 Grad-CAM Results

To better understand how the CNN makes predictions, we use Grad-CAM to visualize the regions of the face that contribute most to the model’s decisions. Figure 1 shows representative Grad-CAM overlays for real, GAN, diffusion, and face-swapped images at convolution layer 4. Across categories, the model is relying on the entire face to classify real vs fake. Because these visualizations are not providing meaningful distinctions within the face, we moved to a more detailed analysis by computing averaged activation maps and overlay maps (fake minus real) across all layers (layer1 to layer4), allowing us to better study how the model’s focus evolves and where subtle differences emerge.

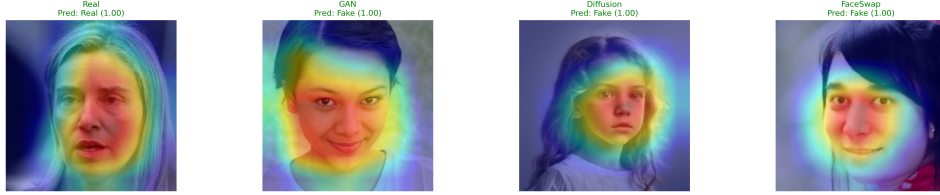


Figure 1: Representative ResNet Grad-CAM examples across real, GAN, diffusion, and face-swapped images.

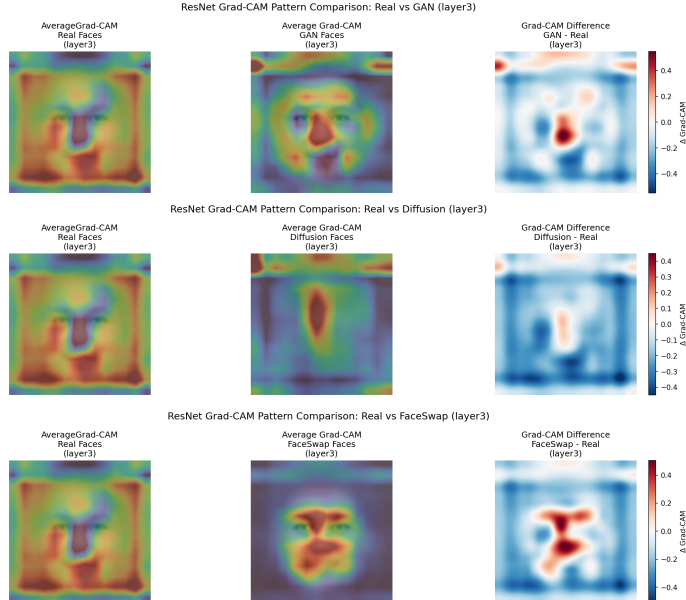


Figure 2: Category-level Grad-CAM comparison (ResNet, layer3).

Early layers (layer1–2) produced noisy, fragmented maps focused on low-level features, while the deepest layer (layer4) yielded more global but overly diffuse activations. Layer3 provided the

best balance, producing structured and interpretable maps with sufficient localization. It captures meaningful facial regions (eyes, nose, mouth) while retaining semantic relevance, so we use it for all main visualizations (Figure 2).

The results show that for GAN images, strong positive responses (red regions driving the prediction toward “fake”) are concentrated in facial features (specifically nose), indicating the model detects localized texture artifacts.

For face-swapped images, there is a clear divide between strong and weak positive activations along the facial boundary, showing that the model focuses on blending and alignment errors.

For diffusion images, unlike the other two types, the results show that the model has difficulty pinpointing specific facial regions that strongly indicate the image is fake (no strong positive signals).

4.3 Vision Transformer (ViT) Results

4.3.1 Grad-CAM Results

Using the same approach as with the CNN model, we apply Grad-CAM to the Vision Transformer (ViT) to better understand the basis of its classifications. Figure 3 presents the Grad-CAM overlays for the same image categories (real, GAN-generated, diffusion-generated, etc.), allowing for a direct comparison between the two architectures.

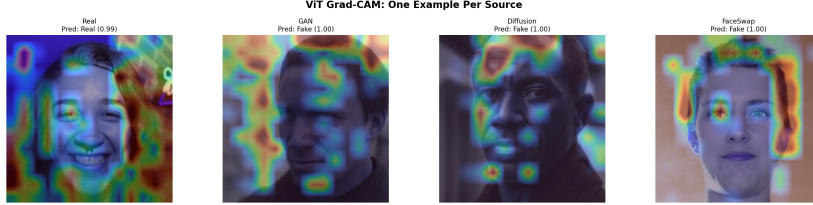


Figure 3: Representative ViT Grad-CAM examples for GAN.

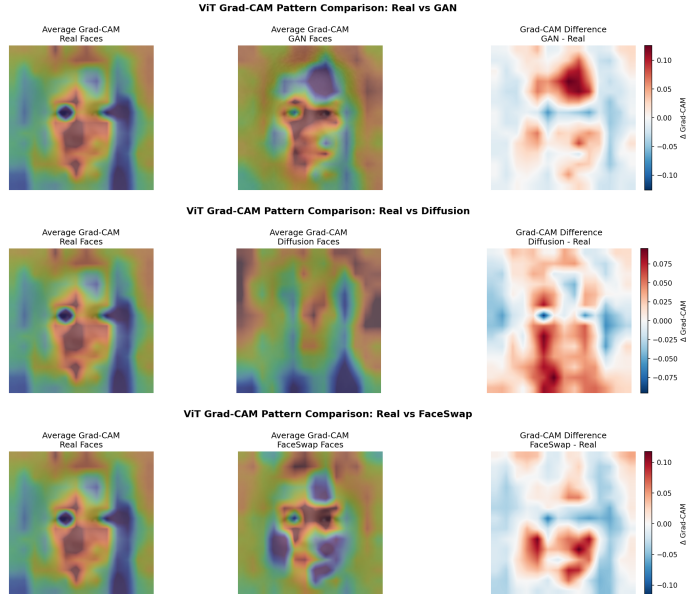


Figure 4: Category-level Grad-CAM overlay analysis comparing real images with GAN, diffusion, and face-swapped images. Each row shows the average activation map for real images, fake images, and their difference.

The difference is reflected in the Grad-CAM visualizations. Compared to the CNN, which places strong emphasis on localized facial regions such as the eyes, nose, and mouth, the ViT exhibits a broader distribution of attention across the entire image. Instead of relying predominantly on isolated facial features, the ViT appears to capture more global characteristics, which may include overall facial symmetry, image-wide texture consistency, structural proportions, and semantic coherence. Although these specific attributes cannot be directly confirmed from the Grad-CAM maps alone, the broader attention patterns suggest that the model depends more heavily on global contextual information when distinguishing between real and synthetic images.

As shown in the right column of Figure 4, the model consistently concentrates its attention on the eye region when classifying real images. In contrast, for synthetic images, stronger attention is distributed across the broader facial region rather than being confined primarily to the eyes. This suggests that, while the eyes remain a critical indicator of authenticity, even highly realistic synthetic images struggle to replicate the subtle realism present in this region. At the same time, the model also detects inconsistencies across the rest of the face, likely in skin texture, facial symmetry, and overall structural coherence. These results indicate that synthetic images fail not only in preserving fine-grained details such as the eyes, but also in reproducing the global characteristics that contribute to a naturally realistic appearance.

This contrast further supports findings from prior work suggesting that Vision Transformers are not solely reliant on localized facial features, but instead capture broader global patterns and structural consistency across the entire image.

4.3.2 Attention Rollout Results

Using attention rollout instead of Grad-CAM reveals similar explanation patterns for classification while also highlighting a fundamental distinction between Transformers and CNNs. Unlike CNN-based explanations, which often emphasize localized regions, attention rollout demonstrates the Transformer’s ability to capture more global contextual information across the entire face. This is reflected in the more distributed spread of attention, where red indicates regions contributing toward “fake” classifications and blue indicates regions contributing toward “real” classifications. Although this experiment does not focus on the fine-grained mechanics of attention rollout itself, the visualizations suggest that the model distinguishes real from fake using broader facial characteristics such as symmetry, overall skin texture, wrinkles, and structural consistency rather than relying solely on isolated local artifacts.

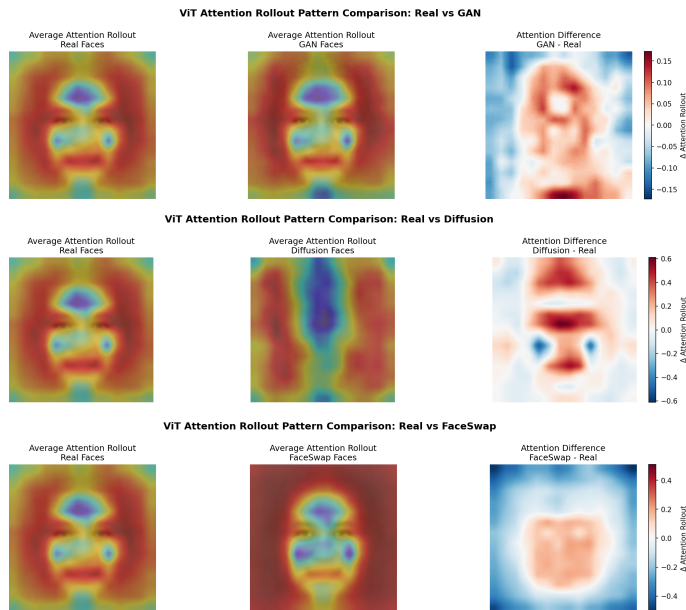


Figure 5: Category-level Attention Rollout overlay analysis comparing real images with GAN, diffusion, and face-swapped images. Each row shows the average activation map for real images, fake images, and their difference.

For GAN-generated faces, the model appears to focus on inconsistencies in global facial realism, which would include unnatural symmetry, overly smooth or artificial skin texture, and subtle structural irregularities.

In diffusion-generated faces, humans can often identify the synthetic appearance through unnatural contours and unrealistic skin texture, and this behavior is reflected in the attention maps, particularly with stronger “fake” attribution concentrated around the cheek regions.

For face-swap manipulations, much of the face and head remains photorealistic because they originate from real images, but belong to different individuals. As a result, the emphasis for real and fake clearly defines the facial boundary regions, where blending inconsistencies and identity mismatches are most apparent (Figure 5). The difference maps clearly highlight these sharp transitions, indicating that the Transformer is effectively capturing the boundary artifacts that characterize face-swap forgeries.

5 Conclusion

From our observations, CNN-based models appear to identify stronger and more localized signals in GAN-generated and face-swap images. Grad-CAM often highlights clear forensic regions such as facial boundaries, blending artifacts, and texture inconsistencies, making the model’s reasoning easier to interpret. For diffusion-generated images, however, Grad-CAM produces weaker and less distinct activation patterns, making it harder to justify the source of classification. This suggests that diffusion-generated faces may contain fewer obvious local artifacts and rely more on subtle global inconsistencies. Vision Transformers provide richer interpretability through both Grad-CAM and attention rollout. These results reinforce the difference between local and global feature learning: CNNs focus more on localized artifacts, while Vision Transformers capture broader structural information such as facial symmetry, skin texture, contour consistency, and overall facial coherence. This helps better explain why artificially generated images often appear fake to the human eye, even when obvious local defects are absent. Overall, CNNs are highly effective for detecting localized manipulations such as GAN artifacts and face swaps, while Vision Transformers offer stronger explanatory power for more globally distributed inconsistencies, particularly in diffusion-generated images.

5.1 Limitation

One limitation of this project is the quality and consistency of the dataset. The portrait backgrounds vary significantly across samples, which introduces visual noise. As a result, both Grad-CAM and attention rollout sometimes highlight background regions instead of focusing entirely on facial features. This suggests that part of the model’s learned representation may be influenced by background artifacts rather than purely facial forensic cues. We recognize this as a source of noise that affects the interpretability of the visualizations and the reliability of the averaged results.

5.2 Future Work

A natural next step is to reverse engineer the generation process itself. Instead of only studying how models detect fake images, we would like to investigate whether image generators can be prompted to avoid the forensic signals identified by our models. By designing prompts that intentionally reduce artifacts such as unnatural symmetry, unrealistic skin texture, or boundary inconsistencies, we can test whether these changes make generated images harder to classify as fake. This would help determine whether current detection models rely mainly on superficial artifacts or on deeper structural inconsistencies in synthetic image generation.

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